

Project Overview

This interactive HR analytics dashboard was built using Power BI Desktop with the IBM HR Analytics Employee Attrition dataset, comprising 1,470 employee records across three departments. The dashboard is designed to help HR decision makers and business leaders understand workforce trends, identify attrition risks, and evaluate employee satisfaction and compensation patterns across the organisation.

Dataset Notes

Several columns in the dataset use numeric encodings which were interpreted as follows throughout the dashboard:

Education: 1 = Below College, 2 = College, 3 = Bachelor, 4 = Master, 5 = Doctor

Environment Satisfaction, Job Satisfaction, and Relationship Satisfaction: 1 = Low, 2 = Medium, 3 = High, 4 = Very High

Job Involvement: 1 = Low, 2 = Medium, 3 = High, 4 = Very High

Performance Rating: 1 = Low, 2 = Good, 3 = Excellent, 4 = Outstanding

Work Life Balance: 1 = Bad, 2 = Good, 3 = Better, 4 = Best

Pages and Key Insights

Overview presents a high level snapshot of the workforce, including total headcount, attrition count (237), average satisfaction score (2.73, equivalent to Medium to High), and average monthly income (6.50K). Pie & Donut charts visualise attrition rate (16.12%), promotion rate (4.42%), and gender distribution, while a clustered column chart breaks down age distribution by gender across the full employee age range.

Performance and Compensation explores the relationship between employee tenure, education level, and earnings. A treemap compares average monthly income across departments, a bar chart reveals a clear income progression from Below College through to Doctoral level education, and two scatter plots examine how years at the company and monthly income relate to performance ratings. The majority of employees are rated either Excellent (3) or Outstanding (4), with no strong linear relationship observed between tenure and performance.

Employee Satisfaction and Engagement analyses workforce wellbeing through work life balance scores (most employees rating Good to Better), job involvement distribution (the majority scoring High involvement at level 3), overtime impact on satisfaction, and average

satisfaction scores by department. The page reveals that employees working overtime report marginally higher satisfaction scores, and that Sales leads department level satisfaction.

Attrition and Work Patterns investigates attrition behaviour in the context of commuting distance and gender, using donut charts and a scatter plot to surface patterns between distance from home and likelihood of attrition.

Technical Details

- Built in Power BI Desktop with four report pages and cross page slicer navigation.
- DAX calculated columns used to convert attrition and promotion fields from categorical to numeric for aggregation.
- Slicers for Gender and Department filter all visuals simultaneously across each page.
- Bookmark navigation buttons on the sidebar enable seamless page switching.

Tools Used: Power BI Desktop, DAX, IBM HR Analytics Dataset.